J Solution Chem (2010) 39: 1454–1473
DOI 10.1007/s10953-010-9601-2

Thermodynamics of the Interaction of a Homologous Series of Amino Acids with Sorbitol

Niki S. Jha · Nand Kishore

Received: 9 February 2010 / Accepted: 28 April 2010 / Published online: 7 October 2010
© Springer Science+Business Media, LLC 2010

Abstract The apparent molar volumes $V_{2,\phi}$, apparent molar isentropic compressibilities $K_{S,2,\phi}$, and enthalpies of dilution of aqueous glycine, alanine, α-amino butyric acid, valine, and leucine have been determined in aqueous 1.0 and 2.0 mol·dm^{-3} sorbitol solutions at 298.15 K. These data have been used to calculate the infinite dilution standard partial molar volumes $V^0_{2,m}$, partial molar isentropic compressibilities $K^0_{S,2,m}$, and enthalpies of dilution $\Delta_{\text{dil}}H^0$ of the amino acids in aqueous sorbitol, along with the standard partial molar quantities of transfer of the amino acids from water to aqueous sorbitol. The linear correlation of $V^0_{2,m}$ for this homologous series of amino acids has been utilized to calculate the contribution to V^0_2 of the charged end groups (NH^+_3, COO^-), the CH_2 group, and other alkyl chains of the amino acids. The results for the standard partial molar volumes of transfer, compressibilites and enthalpies of dilution from water to aqueous sorbitol solutions have been correlated and interpreted in terms of ion–polar, ion–hydrophobic, and hydrophobic–hydrophobic group interactions. A comparison of these thermodynamic properties of transfer suggest that an enhancement of the hydrophilic/polar group interactions is operating in ternary systems of amino acid, sorbitol, and water.

Keywords Partial molar volume · Partial molar isentropic compressibility · Enthalpy of dilution · α-Amino acids · Hydration number · Sorbitol

1 Introduction

Proteins need to maintain their native folded structures to properly function under physiological conditions. However, most proteins are sensitive to changes in cellular and environmental conditions including temperature, pressure, and the presence of salts and other solutes. Therefore, under physiological conditions, proteins require counter-balancing of any significant perturbations in these thermodynamic conditions to avoid significant changes in their secondary and tertiary structures. Proteins that fail to adapt to such conditions may undergo

N.S. Jha · N. Kishore (✉)
Department of Chemistry, Indian Institute of Technology Bombay, Powai, Mumbai 400076, India
e-mail: nandk@chem.iitb.ac.in

a partial or complete loss of their functional activity [1]. In order to adapt to such perturbations from extreme conditions, nature has created certain mechanisms such as the accumulation of small organic solutes, also known as "osmolytes". Osmolytes are small organic salts accumulated at high concentrations by cells or tissues in response to osmotic stress. Osmolytes increase the amount of folded proteins and provide protection against denaturing stress.

The mechanism by which osmolytes aid in assuring protein stability is believed to be a solvent-oriented process whereby protein folding is facilitated by the preferential ordering of solvent molecules. However the exact mechanism of stabilization has been a subject of much discussion. These discussions have led to the conclusion that the macro effects induced in the proteins by these cosolutes are a result of their preferential interaction with the proteins [2]. Preferential interaction is a thermodynamic measure of how solute and solvent redistribute around a protein in solution. After the protein molecule is put into this solution, the molecules of solute will redistribute around the protein in a ratio that is generally not the same as their ratio in bulk solution. If the concentration of water is enriched near the surface relative to its composition in the bulk solution, there is, by definition, a preferential interaction between water and the protein surface. This condition is called preferential hydration. If this occurs, then it stands to reason that if water is enriched near the surface then the solutes will be in deficit, so preferential hydration implies preferential exclusion of solutes [3].

Polyols are known to stabilize proteins, and sorbitol is widely used as a stabilizing agent in the food industry as well as in pharmaceutical formulations since it is harmless upon ingestion [4–6]. In the literature it has been proposed that sorbitol and polyol-like compounds exhibit selective repulsion or attraction with certain loci on the protein surface. Generally, these types of molecules are repelled by certain groups on the protein leading to the formation of a hydration layer. The predominant cause of this process is that contacts between water and nonpolar sites on the protein are more favorable than those between these sites and polyol-like additives [7].

However, several details in the mode of interaction of sorbitol with the proteins remain unanswered. The present study attempts to understand quantitatively and qualitatively the effect of preferential interactions of some building blocks of proteins with sorbitol, using a combination of densimetry, compressibility and isothermal titration calorimetry. To the best our knowledge, no report is available in the literature on the volumetric properties of these amino acids in aqueous sorbitol solutions.

2 Experimental

Glycine, L-alanine, DL-α-amino-n-butyric acid, L-valine and L-leucine were procured from Sigma-Aldrich Co., USA. D-Sorbitol was of extra pure analytical reagent grade purchased from Sisco Research Laboratories, India. The purity of these chemicals as reported by the vendors is listed in Table 1. All five amino acids under investigation were dried over P_2O_5 in a vacuum desiccator prior to use. Corrections were made for the moisture content in these amino acids wherever necessary. The water used for the solution preparations was double distilled, deionized by an ion-exchange column (Cole Parmer), followed by degassing. All weights were determined on a Sartorius BP211D digital balance, which had a readability of ± 0.01 mg.

Table 1 Compounds used in this study with their empirical formula, molecular weight (M_r), source, and their mole fraction purity as reported by the vendors

Compound	Formula	M_r	Source[a]	Purity
Glycine	$C_2H_5NO_2$	75.07	S	>0.99
L-Alanine	$C_3H_7NO_2$	89.09	S	>0.99
DL-α-Amino-n-butyric acid	$C_4H_9NO_2$	103.1	S	>0.99
L-Valine	$C_5H_{11}NO_2$	117.1	S	>0.98
L-Leucine	$C_6H_{13}NO_2$	131.2	S	>0.98
D-Sorbitol	$C_6H_{14}O_6$	182.2	SRL	>0.98

[a]Sources: S = Sigma Aldrich Company, USA; SRL = Sisco Research Laboratory, India

2.1 Density and Sound Velocity Measurements

The density and speed of sound measurements were done on a DSA 5000 digital density and sound velocity analyzer purchased from Anton Paar GmbH, Austria. The calibration of the densimeter was performed daily by measuring values of the apparent molar volume and adiabatic compressibility of aqueous NaCl solutions at different molalities and comparing these results with literature values [8–10], which are in excellent agreement. The values of density and sound velocity were observed to be reproducible within 3×10^{-6} g·cm^{-3} and 0.1 m·s^{-1}, respectively.

Values of the apparent molar volume, $V_{2,\phi}$, and the apparent molar adiabatic compressibility, $K_{S,2,\phi}$, of aqueous solutions were determined, respectively, from the measured density ρ and adiabatic compressibility κ_S by using the following equations

$$V_{2,\phi} = \frac{M}{\rho} - \frac{1000(\rho - \rho_0)}{m\rho\rho_0} \tag{1}$$

and

$$K_{S,2,\phi} = \frac{\kappa_s M}{\rho} - \frac{1000(\kappa_{s,0}\rho - \kappa_S\rho_0)}{m\rho\rho_0} \tag{2}$$

where M is the molar mass of the solute in g·mol^{-1}, m is the molality in mol·kg^{-1} of the amino acid in sorbitol–water mixtures, ρ and ρ_0 are the densities of the amino acid–sugar–water ternary system and of the reference solvent (aqueous sorbitol at the desired molality), respectively, and $\kappa_{S,0}$ is the compressibility of the respective solvent.

The adiabatic compressibility, κ_S, required to evaluate $K_{S,2,\phi}$, was calculated from the density and sound velocity (u) data using the following equation

$$\kappa_S = (\rho u^2)^{-1} \tag{3}$$

2.2 Isothermal Titration Calorimetry (ITC)

ITC measurements were performed at 298.15 K with a titration microcalorimeter (VP-ITC, Microcal Northampton, MA, USA). All of the solutions were thoroughly degassed using a Thermovac sample degassing and thermostat unit provided with the instrument. The reference cell was filled with the degassed and deionized water. The sample cell of volume 1.4206 cm^3 was filled with aqueous sorbitol solution. Titrations were carried out using a 0.25 cm^3 syringe filled with the amino acid solution of interest, with stirring at a speed of 300 rpm. The amino acid solution was added sequentially in 10 µL aliquots (for a total of 25 injections, 20 s duration each) at 4 min intervals. The heat released or absorbed upon

each injection was measured, and the data were plotted as integrated quantities. Enthalpies of dilution were determined by titrating water against the aqueous 1 mol·dm^{-3} sorbitol solution, and corrections were applied to the main sorbitol–amino acid heat profiles. Titrations were performed with 0.2 mol·dm^{-3} glycine, alanine, α-amino-n-butyric acid, and of 0.05 mol·dm^{-3} valine and leucine, with deionized water and 1 mol·dm^{-3} aqueous sorbitol. A typical ITC experiment provides a set of values of the heat liberated at different molalities of the amino acids. The limiting enthalpy of dilution ($\Delta_{\mathrm{dil}}H^0$) of the respective amino acid solution was calculated by fitting the values of the measured heat (q) to the following equation,

$$q = \Delta_{\mathrm{dil}}H^0 + mS_V \tag{4}$$

Here m is the molality of the amino acid in the solution and S_V is the empirical slope.

Considering the errors present in the mass determination, sample impurities, and instrumental reproducibility, we assign a maximum uncertainty of 2% to the measured values of q. The value of limiting enthalpies of transfer ($\Delta_{\mathrm{tr}}\Delta H^0$) of aqueous amino acids from water to aqueous sorbitol solutions were calculated from the difference in the values of $\Delta_{\mathrm{dil}}H^0$ in aqueous sorbitol and in water by using following equation:

$$\Delta_{\mathrm{tr}}\Delta H^0[\text{water to sorbitol(aq)}] = \Delta_{\mathrm{dil}}H^0[\text{sorbitol(aq)}] - \Delta_{\mathrm{dil}}H^0[\text{water}] \tag{5}$$

3 Results and Discussion

3.1 Volumetric Properties of Amino Acids in Aqueous Sorbitol Solutions

3.1.1 Partial Molar Volume

Results of the density measurements of amino acids in aqueous sorbitol solutions at 298.15 K are given in Table 2. In the cases of valine and leucine, where the molality dependence of $V_{2,\phi}$ was found to be either negligible or having no definite trend, the value of the standard partial molar volume at infinite dilution $V^0_{2,m}$ was evaluated by averaging all of the data points. In all other cases, the value of the $V^0_{2,m}$ was obtained by a least-square fitting to the following equation:

$$V_{2,\phi} = V^0_{2,m} + S_V m \tag{6}$$

Here S_V, is a volumetric virial coefficient that characterizes the pair wise interaction of solvated solute species in solution [11, 12]. The values of the $V^0_{2,m}$ along with their corresponding S_V are given in Table 3. The sign of S_V is determined by the nature of the interaction between the solute species. For the zwitterionic amino acids, the positive values of S_V suggest that the pair wise interactions are dominated by interactions of the charged functional groups. The variation of S_V values with the side chain position indicates that the methyl group modulates the interaction of the charged end groups in pair wise interactions.

The values of $V^0_{2,m}$ are positive for all of the studied amino acids in aqueous sorbitol solutions and increase with increasing molality of sorbitol. For glycine, alanine and α-amino-n-butyric acid, the values of $V^0_{2,m}$ are higher in aqueous sorbitol solutions. However, for L-valine, the value of $V^0_{2,m}$ remains nearly the same, and for L-leucine the values of $V^0_{2,m}$ are smaller in aqueous sorbitol compared to those in water. The values of $V^0_{2,m}$ show a linear variation with the number of carbon atoms in the alkyl chain of the amino acids, with an

Table 2 Density, ρ, apparent molar volume, $V_{2,\phi}$, velocity of sound, u, and apparent molar adiabatic compressibility, $K_{S,2,\phi}$, of aqueous sorbitol in different amino acids

m (mol·kg^{-1})	ρ (g·cm^{-3})	u (m·s^{-1})	$V_{2,\phi}$ (cm^3·mol^{-1})	k_s (10^9 hPa^{-1})	$K_{S,2,\phi}$ (10^{-7} cm^3·mol^{-1}·hPa^{-1})
Solvent: 1.0 mol·dm^{-3} sorbitol					
Glycine					
0.00000	1.060112	1569.95		38.2716	
0.09497	1.062959	1575.43	44.02	37.9040	–19.82
0.10624	1.063291	1576.01	44.06	37.8604	–19.83
0.19854	1.065955	1581.05	44.38	37.5297	–18.59
0.21579	1.066444	1582.06	44.44	37.4641	–18.65
0.24405	1.067255	1583.62	44.47	37.3620	–18.54
0.29428	1.068683	1586.21	44.53	37.1902	–18.10
0.35699	1.070390	1589.62	44.76	36.9716	–17.80
0.42324	1.072194	1591.55	44.90	36.7496	–17.42
0.45418	1.073034	1594.81	44.95	36.6408	–17.40
0.49265	1.074058	1596.81	45.02	36.5145	–17.20
L-Alanine					
0.00000	1.059490	1569.54		38.3141	
0.16224	1.063622	1580.57	61.16	37.6313	–16.79
0.20321	1.064643	1583.38	61.20	37.4652	–16.50
0.24300	1.065629	1585.93	61.48	37.3102	–16.15
0.33233	1.067730	1592.04	61.52	36.9517	–15.96
0.36303	1.068458	1594.10	61.56	36.8305	–15.90
0.36642	1.068536	1594.36	61.57	36.8168	–16.00
0.42074	1.069794	1597.07	61.67	36.6123	–15.60
0.46615	1.070838	1601.89	61.74	36.4420	–15.40
0.65598	1.075118	1614.44	61.95	35.7748	–14.40
DL-α-Amino-n-butyric acid					
0.00000	1.059637	1569.74		38.2990	
0.12958	1.062696	1580.72	76.07	37.6598	–17.91
0.15271	1.063231	1582.71	76.10	37.5463	–17.95
0.19200	1.064141	1585.96	76.10	37.3606	–17.70
0.21765	1.064730	1588.00	76.11	37.2441	–17.40
0.30036	1.066607	1594.61	76.15	37.8709	–16.80
0.34491	1.067597	1598.12	76.19	36.6749	–16.50
0.40072	1.068823	1602.69	76.24	36.4242	–16.39
0.51139	1.071223	1611.63	76.29	35.9408	–16.10
0.58206	1.072711	1617.43	76.37	35.6341	–16.00
0.60765	1.073219	1618.86	76.43	35.5542	–15.46
L-Valine					
0.00000	1.059820	1569.85		38.2870	
0.04214	1.060758	1574.22	90.60	38.0401	–20.40
0.06475	1.061244	1576.55	90.79	37.9118	–20.24

Table 2 (*Continued*)

m (mol·kg^{-1})	ρ (g·cm^{-3})	u (m·s^{-1})	$V_{2,\phi}$ (cm^3·mol^{-1})	k_s (10^9 hPa^{-1})	$K_{S,2,\phi}$ (10^{-7} cm^3·mol^{-1}·hPa^{-1})
0.07414	1.061448	1577.57	90.80	37.8558	–20.50
0.07976	1.061561	1578.03	90.91	37.8289	–19.80
0.08452	1.061655	1578.40	91.01	37.8077	–19.10
0.08990	1.061767	1579.08	91.04	37.7712	–19.74
0.09225	1.061816	1579.46	91.06	37.7514	–20.40
0.09657	1.061905	1579.86	91.09	37.7290	–20.15
0.10349	1.062048	1580.55	91.13	37.6909	–20.00
0.13374	1.062688	1583.77	91.15	37.5155	–20.23
L-Leucine					
0.00000	1.060182	1570.39		38.2476	
0.04011	1.060974	1576.21	106.11	37.9374	–21.40
0.04808	1.061123	1577.29	106.20	37.8797	–20.89
0.04935	1.061146	1577.33	106.28	37.8777	–19.97
0.06029	1.061339	1578.82	106.56	37.7988	–19.57
0.06181	1.061365	1578.77	106.60	37.8005	–18.25
0.06398	1.061397	1579.20	106.73	37.7571	–18.85
0.06702	1.061450	1579.61	106.79	37.7557	–18.76
0.08025	1.061759	1581.33	106.96	37.6643	–18.69
Solvent: 2.0 mol·dm^{-3} sorbitol					
Glycine					
0.00000	1.120192	1652.62		32.6859	
0.10173	1.122854	1657.89	46.05	32.4012	–10.06
0.13299	1.123659	1659.85	46.10	32.3018	–10.89
0.17237	1.124661	1662.01	46.17	32.1892	–10.86
0.22923	1.126091	1665.14	46.26	32.0275	–10.82
0.25662	1.126769	1666.44	46.32	31.9582	–10.51
0.29791	1.127774	1668.99	46.42	31.8323	–10.80
0.36605	1.129421	1672.85	46.54	31.6392	–10.80
0.48932	1.132367	1679.43	46.68	31.3103	–10.48
0.52299	1.132771	1682.22	46.70	31.1954	–10.68
0.52362	1.133113	1682.22	46.81	31.1861	–10.97
L-Alanine					
0.00000	1.119630	1651.61		32.7423	
0.11239	1.122006	1659.18	62.57	32.3756	–8.98
0.22406	1.124279	1666.83	62.76	32.0140	–8.96
0.26763	1.125136	1669.84	62.85	31.8745	–8.95
0.28563	1.125482	1671.06	62.90	31.8183	–8.90
0.35641	1.126857	1675.68	62.99	31.6044	–8.63
0.38040	1.127321	1677.36	63.01	31.5279	–8.67
0.38932	1.127468	1677.88	63.07	31.5044	–8.55
0.43504	1.128345	1680.89	63.10	31.3672	–8.46

Table 2 (*Continued*)

m (mol·kg^{-1})	ρ (g·cm^{-3})	u (m·s^{-1})	$V_{2,\phi}$ (cm^3·mol^{-1})	k_s (10^9 hPa^{-1})	$K_{S,2,\phi}$ (10^{-7} cm^3·mol^{-1}·hPa^{-1})
0.45189	1.128644	1681.90	63.15	31.3212	–8.33
0.46333	1.128859	1682.54	63.16	31.2917	–8.22
DL-α-Amino butyric acid					
0.00000	1.120863	1653.12		32.6466	
0.10501	1.122827	1661.90	76.98	32.2458	–9.23
0.19141	1.124371	1669.18	77.17	31.9212	–9.18
0.21790	1.124829	1671.39	77.24	31.8241	–9.10
0.26196	1.125603	1675.05	77.27	31.6632	–9.03
0.26796	1.125703	1675.49	77.30	31.6439	–8.93
0.30530	1.126345	1678.10	77.33	31.5277	–8.32
0.39238	1.127852	1685.60	77.34	31.2058	–8.63
0.42232	1.128348	1688.10	77.38	31.0997	–8.62
0.42718	1.128414	1688.41	77.41	31.0867	–8.52
0.50793	1.129739	1695.03	77.46	31.0815	–8.16
L-Valine					
0.00000	1.120556	1653.21		32.6500	
0.05741	1.121530	1658.96	90.91	32.3978	–9.74
0.06220	1.121603	1659.41	91.01	32.3782	–9.52
0.07698	1.121838	1660.89	91.17	32.3137	–9.53
0.10361	1.122267	1663.69	91.21	32.1926	–10.03
0.10535	1.122294	1663.85	91.22	32.1858	–9.96
0.11918	1.122515	1665.09	91.25	32.1314	–9.51
0.12977	1.122682	1666.33	91.28	32.0787	–10.00
0.14235	1.122866	1667.44	91.39	32.0310	–9.53
0.14764	1.122941	1668.01	91.44	31.0068	–9.61
L-Leucine					
0.00000	1.120664	1653.31		32.6449	
0.02929	1.121061	1656.83	106.25	32.4941	–10.94
0.03306	1.121102	1657.25	106.48	32.4771	–10.42
0.04283	1.121277	1658.32	106.51	32.4301	–10.21
0.04599	1.121322	1658.49	106.52	32.4221	–9.42
0.04629	1.121324	1658.53	106.55	32.4206	–8.68
0.05210	1.121399	1659.18	106.66	32.3928	–8.62
0.05300	1.121411	1659.28	106.67	32.3886	–8.60
0.06799	1.121522	1661.09	106.95	32.3148	–8.47
0.06950	1.121539	1661.26	106.97	32.3077	–8.44

average regression coefficient of 0.999. This linear variation of $V^0_{2,m}$ is represented by the following equation [13–15],

$$V^0_{2,m} = V^0_{2,m}(\mathrm{NH_3^+}, \mathrm{COO^-}) + n_c V^0_{2,m}(\mathrm{CH_2}) \tag{7}$$

Table 3 Volumetric parameters of amino acids and peptides in aqueous sorbitol at 298.15 K

Amino acids	Parameter	Water	Solvent	
			1.0 mol dm^{-3} aq. sorbitol	2.0 mol dm^{-3} aq. sorbitol
Glycine	$V^0_{2,m}$	43.46 (0.01)	44.83 (0.04)	44.88 (0.02)
	S_V	0.86 (0.09)	2.47	1.67 (0.06)
	R	0.97	0.99	0.99
Alanine	$V^0_{2,m}$	60.43 (0.04)	60.97 (0.070)	62.40 (0.017)
	S_V	0.73	1.58	1.64
	R	0.99	0.96	0.99
α-Amino-n-butyric acid	$V^0_{2,m}$	75.51 (0.00)	75.96 (0.024)	77.07 (0.033)
	S_V		0.691	0.74 (0.099)
	R		0.98	0.94
Valine	$V^0_{2,m}$	90.26 (0.14)	90.26 (0.04)	90.77 (0.05)
	S_V		8.34 (0.55)	4.2 (0.51)
	R		0.97	0.95
Leucine	$V^0_{2,m}$	107.72 (0.24)	105.14 (0.11)	105.75 (0.071)
	S_V		23.58 (1.90)	16.26 (1.41)
	R		0.98	0.97

[a]Entries in the parentheses are standard deviations. Units: $V^0_{2,m}$ (cm^3·mol^{-1}); S_V (cm^3·mol^{-2}·kg). R: Regression coefficient of the linear fit

Table 4 Contribution of the zwitterionic group (NH_3^+, COO^-), CH_2 group and the other alkylchains to the infinite dilution apparent molar volume in aqueous sorbitol at 298.15 K

Group	$V^0_{2,m}$ (cm^3·mol^{-1})		
	Water	Sorbitol	
		1.0 mol·kg^{-1}	2.0 mol·kg^{-1}
NH_3^+, COO^-	27.68 (1.12)	29.76 (1.04)	29.92 (1.24)
CH_2-	15.91 (0.33)	15.16 (0.31)	15.36 (0.37)
CH_3CH-	31.82 (0.40)	30.32 (0.62)	30.69 (0.74)
CH_3CH_2CH-	47.73 (0.40)	45.47 (0.62)	46.05 (0.74)
CH_3CH_3CHCH-	63.64 (0.40)	60.63 (0.62)	61.40 (0.74)
$CH_3CH_3CHCH_2CH$-	79.55 (0.40)	75.80 (0.62)	76.75 (0.74)

Values in the parentheses are standard deviation values. Average correlation coefficient: 0.999

where n_c is the number of carbon atoms in the alkyl chain of the α-amino acids; $V^0_{2,m}(NH_3^+, COO^-)$ and $V^0_{2,m}(CH_2)$ are the zwitterionic end groups and methylene group contributions to $V^0_{2,m}$, respectively. The values of $V^0_{2,m}(NH_3^+, COO^-)$ and $V^0_{2,m}(CH_2)$, calculated by least-squares regression analysis, are listed in Table 4. Since the alkyl chains of the homologous series of α-amino acids studied in this work are CH_2- (glycine), CH_3CH- (alanine), CH_3CH_2CH- (α-amino-n-butyric acid), CH_3CH_3CHCH- (valine), and $CH_3CH_3CHCH_2CH$- (leucine), the value of $V^0_{2,m}(CH_2)$ obtained by this procedure characterizes the mean contribution of the CH- and CH_3- groups to $V^0_{2,m}$. As suggested by Hakin and coworkers [16, 17], the contribution of the other alkyl chains of the amino acids reported

in Table 4 were calculated as follows:

$$V^0_{2,m}(\mathrm{CH_3}) = 1.5V^0_{2,m}(\mathrm{CH_2}) \tag{8}$$

$$V^0_{2,m}(\mathrm{CH}) = 0.5V^0_{2,m}(\mathrm{CH_2}) \tag{9}$$

The values of $V^0_2(\mathrm{CH_2})$, 15.16 ± 0.31 cm^3·mol^{-1} and 15.36 ± 0.37 cm^3·mol^{-1}, in 1.0 and in 2.0 mol·dm^{-3} aqueous sorbitol are nearly the same. For the same set of amino acids, the values of $V^0_2(\mathrm{CH_2})$ are (15.2 ± 0.2) cm^3·mol^{-1} in aqueous sodium acetate [15], 15.3 and 15.6 cm^3·mol^{-1} in aqueous KSCN [14], (16.08 ± 0.68) cm^3·mol^{-1}[18] in aqueous sodium dodecyl sulfate (SDS), and (15.83 ± 0.29) aqueous cetyl trimethyl ammonium bromide (CTAB) solutions at 298.15 K [12]. The values of $V^0_{2,m}(\mathrm{CH_2})$ observed in aqueous sorbitol are also not very different from that in water (15.91 ± 0.33 cm^3·mol^{-1}). Larger values of $V^0_{2,m}(\mathrm{NH_3^+}, \mathrm{COO^-})$ compared to $V^0_{2,m}(\mathrm{CH_2})$ indicate that the interactions of the hydroxyl groups of sorbitol with the zwitterionic groups of the amino acids dominate when compared to those from the hydrophobic groups of the amino acids. In addition, the difference in the molar masses of the $(\mathrm{NH_3^+}, \mathrm{COO^-})$ and $\mathrm{CH_2}$ groups also accounts for the larger $V^0_{2,m}$ values of the former. The number of water molecules hydrated to the amino acids, N_w, was calculated from the values of the measured standard partial molar volumes by using the method described immediately below.

The values of $V^0_{2,m}$ for the amino acids can be expressed [19] by

$$V^0_{2,m} = V^0_{2,m}(\mathrm{int}) + V^0_{2,m}(\mathrm{elect}) \tag{10}$$

where $V^0_{2,m}(\mathrm{int})$ is the intrinsic partial molar volume of the amino acid and $V^0_{2,m}(\mathrm{elect})$ is the electrostriction partial molar volume due to hydration of the amino acids. The term $V^0_{2,m}(\mathrm{int})$ can be further divided into two terms, one for the van der Waals volume and the other for volume resulting from packing effects. Millero et al. [20] have estimated values of $V^0_{2,m}(\mathrm{int})$ for amino acids from their molar crystal volumes by using the relationship:

$$V^0_{2,m}(\mathrm{int}) = \left(\frac{0.7}{0.634}\right) V^0_{2,m}(\mathrm{cryst}) \tag{11}$$

where 0.7 is the packing density for the molecule in an organic crystal and 0.634 is the packing density for randomly packed spheres. The molar volume of the crystal was calculated from the crystal densities of the amino acids reported by Berlin and Pallansh [21] at $T = 298.15$ K. The values of $V^0_{2,m}(\mathrm{elect})$ were obtained from the experimentally measured values of $V^0_{2,m}$ using Eq. 10

Further, the decrease in the volume due to electrostriction can be related [20] to the hydration number N_w, of the amino acids as

$$N_w = \frac{V^0_2(\mathrm{elect})}{V^0_E - V^0_B} \tag{12}$$

Here V^0_E is the molar volume of the electrostricted water and V^0_B is the molar volume of bulk water at 298.15 K. This model assumes that for every water molecule moved from the bulk phase to the region near the amino acid, the volume is decreased by $(V^0_E - V^0_B)$. Using $(V^0_E - V^0_B) \approx -3.0$ cm^3·mol^{-1} [20] for electrolytes at 298.15 K, the hydration numbers in Table 5 have been calculated. The calculated values of N_w for the amino acids in aqueous sorbitol are observed to vary in the following order:

$$N_w(\mathrm{leucine}) > N_w(\mathrm{valine}) > N_w(\mathrm{alanine}) > N_w(\mathrm{glycine})$$

Table 5 The number of water molecules hydrated (N_w) to the amino acids in water and in aqueous sorbitol at 298.15 K

Cosolvent	m (mol·kg^{-1})	N_w			
		Glycine	Alanine	Valine	Leucine
Water	0.0	2.9	3.8	3.9	5.5
Sorbitol	1.0	2.7	3.6	3.9	6.4
Sorbitol	2.0	2.0	3.2	3.7	6.1

Table 6 Transfer volumes $\Delta_{tr}V^0_{2,m}$ of amino acids from water to aqueous sorbitol at 298.15 K

Amino acids	$V^0_{2,m}$ (cm^3·mol^{-1}), water	$\Delta_{tr}V^0_{2,m}$ (cm^3·mol^{-1}) sorbitol	
		1 mol·kg^{-1}	2 mol·kg^{-1}
Glycine	43.14 (0.06)	1.37 (0.72)	1.42 (0.054)
Alanine	60.43 (0.04)	0.54 (0.081)	1.97 (0.043)
α-Amino butyric acid	75.51 (0.00)	0.45 (0.024)	1.54 (0.029)
Valine	90.39 (0.14)	0.00 (0.46)	0.57 (0.158)
Leucine	107.72 (0.24)	−2.58 (0.26)	−1.90 (0.25)

This same order is maintained at both concentrations of sorbitol studied. In general, with an increase in the concentration of sorbitol, the N_w values exhibit a slight decrease indicating reduction in the degree of electrostriction.

3.1.2 Standard Partial Molar Volumes of Transfer of Amino Acids from Water to Aqueous Sorbitol

Limiting thermodynamic properties of transfer yield qualitative and quantitative information regarding the interactions of a cosolvent and a solute without the need to take into account the effects of solute–solute interactions. Values of standard partial molar volume of transfer ($\Delta_{tr}V^0_{2,m}$) for the amino acids from water to aqueous sorbitol solutions presented in Table 6 were calculated as follows:

$$\Delta_{tr}V^0_{2,m}[\text{water to sorbitol(aq)}] = V^0_{2,m}[\text{sorbitol(aq)}] - V^0_{2,m}[\text{water}] \tag{13}$$

The uncertainties in the values of volume of transfer were calculated as the root-mean-squares of the uncertainties associated with their respective apparent molar properties at infinite dilution. Values of $\Delta_{tr}V^0_{2,m}$ for transfer of the amino acids from water to aqueous sorbitol at both sorbitol concentrations were found to be positive, except for leucine.

The observed values of $\Delta_{tr}V^0_{2,m}$ of amino acids from water to aqueous sorbitol follow the following order:

$$\text{glycine} > \text{alanine} > \alpha\text{-amino butyric acid} > \text{valine} > \text{leucine}.$$

Glycine, the smallest amongst the five amino acids studied, has the highest value of $\Delta_{tr}V^0_{2,m}$ from water to aqueous sorbitol solution. For these amino acids, the value of $\Delta_{tr}V^0_{2,m}$ becomes more positive on being transferred from 1.0 mol·kg^{-1} to 2.0 mol·kg^{-1} aqueous sorbitol. Only leucine shows a negative value of $\Delta_{tr}V^0_{2,m}$.

Franks et al. [19] explained the standard partial molar volume of a nonelectrolyte as being a combination of the intrinsic volume of the solute and the volume change due to

its interaction with the solvent [22]. The intrinsic volume is considered to consist of two contributions.

$$V_{\text{intrinsic}} = V_{\text{vW}} + V_{\text{void}} \tag{14}$$

where V_{vW} is the volume occupied by the solute due to its van der Waals volume [23] and V_{void} is the volume associated with the voids and the empty spaces present therein [24]. This equation was modified by Shahidi et al. [25] in order to evaluate the contribution of a solute molecule to its standard partial molar volume, as

$$V_{2,m}^{0} = V_{\text{vW}} + V_{\text{void}} - n\sigma_{\text{s}} \tag{15}$$

where σ_{s} is the shrinkage in volume produced by interaction of the hydrogen bonding groups of the solute with water molecules, and n is the number of potential hydrogen bonding sites on the molecule. Thus, $V_{2,m}^{0}$ of an amino acid is given by

$$V_{2,m}^{0} = V_{\text{vW}} + V_{\text{void}} - V_{\text{shrinkage}} \tag{16}$$

If it is assumed that V_{vW} and V_{void} remain appreciably the same in both water and in aqueous sorbitol solutions, the positive volumes of transfer for glycine, alanine, α-amino butyric acid and valine can be rationalized in terms of a decrease in the volume of shrinkage in the presence of the sorbitol molecules in the solutions. Since $V_{2,m}^{0}$ is, by definition, free of solute–solute interactions, it provides information on solute–solvent interactions.

Properties of water molecules in the hydration cosphere depend on the nature of the solute species. The following type of interactions can occur in ternary systems of amino acid, sorbitol and water: (a) ion–hydrophilic interactions between the –OH group of sorbitol and zwitterionic groups of amino acids; (b) hydrophilic–hydrophobic group interactions between –OH of sorbitol and hydrophobic groups of amino acids; and (c) hydrophobic–hydrophobic group interactions between the hydrophobic groups of sorbitol and amino acids.

Taking the cosphere overlap model as a guideline, interactions of type (a) would lead to positive $\Delta_{\text{tr}} V_{2,m}^{0}$ values since they lead to a reduction in the electrostriction effect and the overall water structure is enhanced. On the other hand, interactions of type (b) and (c) would lead to a negative $\Delta_{\text{tr}} V_{2,m}^{0}$ values, because the introduction of the alkyl group provides an additional tendency for hydrophilic–hydrophobic and hydrophobic–hydrophobic groups to interact, leading to a reduction in the structure of water as their cospheres overlap.

Glycine, the smallest amino acid studied, shows the highest positive values of $\Delta_{\text{tr}} V_{2,m}^{0}$. With an increase in the hydrophobic content of the amino acid, the value for $\Delta_{\text{tr}} V_{2,m}^{0}$ become less positive and eventually becomes negative for L-leucine. These results suggest that: type (a) interactions predominate in the case of glycine, whereas (b) and (c) type interactions predominate for leucine. Sorbitol, having six –OH groups, can engage in stronger hydrophilic interactions with the polar groups of the amino acids at higher concentrations, and hence this effect leads to higher positive values of $\Delta_{\text{tr}} V_{2,m}^{0}$ for 2 mol·kg^{-1} compared to 1 mol·kg^{-1} aqueous sorbitol.

3.1.3 Partial Molar Adiabatic Compressibility

Values of apparent molar compressibility $K_{S,2,\phi}$ of amino acids in aqueous sorbitol are given in Table 7. The value of $K_{S,2,\phi}$ at infinite dilution is an important property and provides valuable information on the interactions amongst water, sorbitol and amino acids. The values

Table 7 Isothermal compressibilities, $K_{S,2,\phi}$ of aqueous solution of amino acids in sorbitol at 298.15 K

Amino acid	Parameter	Water	Aqueous sorbitol	
			1.0 mol·kg^{-1}	2.0 mol·kg^{-1}
Glycine	$K^0_{s,2,\phi} \times 10^{-7}$	–27.00 (0.6)	–20.2 (0.15)	–10.68 (0.27)
	$S_{k(s)}$	4.56	0.65	
	R		0.97	
Alanine	$K^0_{s,2,\phi} \times 10^{-7}$	–25.56 (0.6)	–17.4 (0.15)	–9.62 (0.09)
	$S_{k(s)}$	5.01	0.45	2.69
	R		0.98	0.98
α-ABA	$K^0_{s,2,\phi} \times 10^{-7}$	–27.1 (0.6)	–18.47 (0.23)	–9.63 (0.80)
	$S_{k(s)}$		4.82	2.63
	R		0.97	0.97
Valine	$K^0_{s,2,\phi} \times 10^{-7}$	–30.62 (0.6)	–20.05 (0.42)	–9.72 (0.21)
	$S_{k(s)}$	8.43		22.42
	R			0.88
Leucine	$K^0_{s,2,\phi} \times 10^{-7}$	–31.78 (0.6)	–23.99(0.84)	–12.32 (0.66)
	$S_{k(s)}$	13.61	73	61.7
	R		0.91	0.87

Entries in the parentheses are standard deviations. Units: $K_{S,2,\phi}$ (cm^3·mol^{-1}·hPa^{-1}); S_V (cm^3·mol^{-2}·kg·hPa^{-1}). R: Regression coefficient of the linear fit

of standard partial molar adiabatic compressibilities $K^0_{S,2,\phi}$ of amino acids were obtained by fitting the values of $K_{S,2,\phi}$ against m according to the following equations:

$$K_{S,2,\phi} = K^0_{S,2,\phi} + S_K m \tag{17}$$

Here S_K is an empirical constant evaluated from the experimental data.

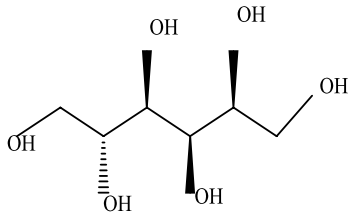
The value of limiting partial molar compressibility of the studied amino acids can be expressed by the simple model given below [20]:

$$K^0_{S,2}(\text{elect}) = K^0_{S,2,m} - K^0_{S,2,m}(\text{int}) \tag{18}$$

where $K^0_{S,2,m}$(int) is the intrinsic partial molar isentropic compressibility of the amino acid and $K^0_{S,2,m}$(elect) is the electrostriction partial molar compressibility due to hydration of the amino acid. Since the value of $K^0_{S,2,m}$(int) is very small, it can be assumed as a first approximation that $K^0_{S,2,m}$(int) is approximately zero [19]. Thus, the value of $K^0_{S,2,m}$ may be thought to represent $K^0_{S,2,m}$(elect). The values of $K^0_{S,2,m}$ for all five studied amino acids are negative in 1.0 and 2.0 mol·kg^{-1} aqueous sorbitol. The negative values arise from interaction of the –OH groups of sorbitol with water due to hydrogen bonding and hence lead to electrostriction. Electrostriction results in a lesser compressibility of the ternary solution of the amino acids–sorbitol–water compared to that of bulk water.

A plot of $K^0_{S,2,m}$ against the number of carbon atoms in the amino acids shows a nonlinear trend, both in water and in aqueous sorbitol solutions (Fig. 2). The value of $K^0_{S,2,m}$ for alanine is less negative compared to that of glycine, indicating relatively less electrostriction of the solvent in the former case. However, with a further increase in the number of carbon atoms in the amino acid, the value of $K^0_{S,2,m}$ become more negative. These results are consistent with those reported in the literature [20]. This trend is also maintained in the presence of aqueous sorbitol. The increases in the negative value of $K^0_{S,2,m}$ for amino acids beyond alanine indicate alteration in the extent of electrostriction as a result of addition hydrophilic

Fig. 1 Molecular structure of sorbitol



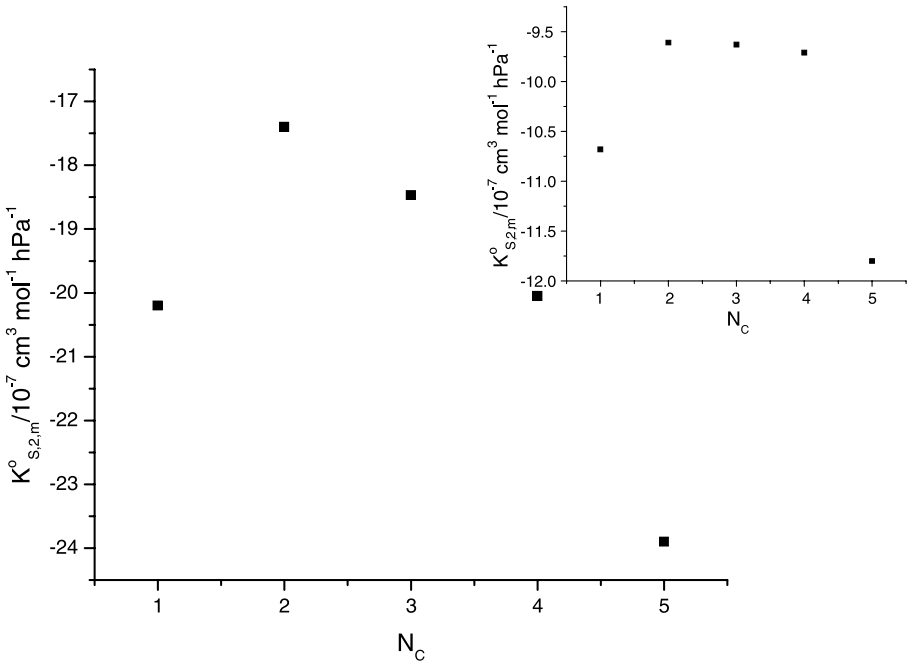


Fig. 2 The partial molar adiabatic compressibility, K_S^0, of the amino acids in (**A**) 1.0 mol·dm^{-3} and (**B**) 2.0 mol·dm^{-3} sorbitol as function of number of carbon atoms in the side chains of the amino acid at 298.15 K

group being present and the larger separation between the zwitterionic end centers in the amino acids (Fig. 1)

As can be seen in Fig. 2, there is a pronounced break at the point corresponding to alanine in both plots. After alanine, the values of $K_{2,\phi}$ decrease with increase in the number of carbon atoms in the amino acid. This observation can be interpreted to mean that only the alkyl group in the β-position interacts with the zwitterionic skeleton of the molecule, perhaps via overlapping hydration shells. After alanine, α-amino butyric acid, each methylene group added is independently hydrated and contributes to the same extent. In the cases of valine and leucine, the β-branched carbon bulk chain is close to the main chain and therefore their hydration shells might well overlap with those of the adjacent backbone region.

The partial molar volume, partial molar adiabatic compressibility, and enthalpy of transfer of some amino acids from water to glucose and sucrose solutions have been reported in the literature [26, 27]. These polyhydroxy compounds are expected to exert interactions similar to those for aqueous sorbitol studied in this work. Except for leucine, where negative

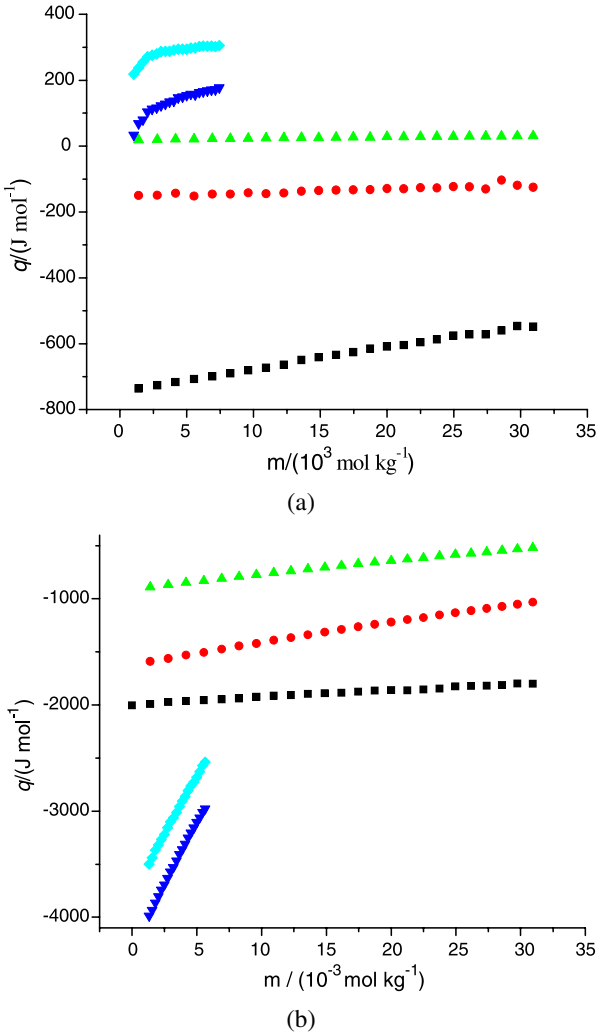


Fig. 3 (**a**) Enthalpy change for the homologous series of amino acids: (■) glycine, (•) alanine, (▲) α-amino butyric acid, (▼) valine, and (◊) leucine upon titration against 1.0 mol·kg^{-1} aqueous sorbitol at 298.15 K. (**b**) Enthalpy change for the homologous series of amino acids: (■) glycine, (•) alanine, (▲) α-amino butyric acid, (▼) valine, and (◊) leucine upon titration against 2.0 mol·kg^{-1} aqueous sorbitol at 298.15 K

volumes of transfer are observed in aqueous sorbitol, the transfer volumes for the remaining four amino acids are nearly the same and follow a similar trend in aqueous glucose and sucrose solutions. The value and sign of $K^0_{S,2,m}$ for glycine in aqueous sorbitol are also comparable with those observed in aqueous glucose solutions [26, 27]. This comparison indicates that the nature of the interactions of the amino acids studied in aqueous sorbitol solutions is similar to those observed in aqueous glucose and sucrose solutions.

3.2 Enthalpies of Dilution of Amino Acids in Aqueous Sorbitol

Values of the heats of dilution for aqueous amino acids were determined by titrating 0.2 mol·dm^{-3} glycine, alanine and α-amino-n-butyric acid, and 0.05 mol·dm^{-3} valine and leucine, in two different concentrations of sorbitol. The enthalpies of dilution in water were taken from the literature [28]. Figures 3a and 3b show the enthalpy change as a function of molality when a solution of amino acid is titrated with aqueous sorbitol. The data obtained

were fitted well by a linear equation and the calculated values of the enthalpies at infinite dilution are reported in Table 8.

The limiting enthalpy of dilution $\Delta_{\text{dil}}H^0$ of glycine in water is endothermic. However, $\Delta_{\text{dil}}H^0$ for glycine becomes more exothermic when diluted in aqueous sorbitol solution after correction for the dilution effect of sorbitol. This exothermicity arises because of hydrophilic interactions between the amino acid and sorbitol. Here again, the magnitude of the exothermicity increases upon increasing the concentration of sorbitol. The limiting enthalpies of dilution of alanine, α-amino-n-butyric acid, and leucine are exothermic in water indicating that net favorable hydrophilic interactions occur for these amino acids with water. However, the corresponding values are exothermic for alanine and endothermic for α-ABA, valine and leucine in 1 mol·kg^{-1} aqueous sorbitol. With further increase in the concentration of the sorbitol in solution, hydrophilic interactions dominate, resulting in negative $\Delta_{\text{dil}}H^0$ values.

The enthalpy of dilution $\Delta_{\text{dil}}H^0$ varies from –742.5 J·K^{-1}·mol^{-1} to –1991.5 J·K^{-1}·mol^{-1} for glycine, when the sorbitol concentration increases from 1.0 mol·dm^{-3} to 2.0 mol·dm^{-3}. A similar trend is observed for the other amino acids when transferred from water to 2 mol·dm^{-3} aqueous sorbitol. The values of $\Delta_{\text{dil}}H^0$ for glycine and alanine are negative. This exothermicity arises from interactions such as H-bonding between the amino acids and water. The positive enthalpies of transfer from water to amino acid solutions for hydrophobic solutes can be rationalized in terms of their structure-breaking effect on water [29]. The negative enthalpies of transfer for sorbitol results from interaction between the amino acids and polyol molecules. The most probable interaction appears to be hydrogen bonding since both amino acid and polyol possess potential sites for hydrogen bonding [30]. According to this view, the size of the exothermicity contribution should decrease with increasing alkyl chain length, because hydrophobic interactions now play a major role. The enthalpy of transfer is taken as a combination of contributions from the nonpolar -CH_2 and polar -OH groups. It appears that with changing the amino acid from glycine to leucine, the contribution of nonpolar interactions (positive $\Delta_{\text{tr}}H$) outweighs the contribution of polar hydroxyl groups (negative $\Delta_{\text{tr}}H$), thus resulting in a positive enthalpy of transfer [31].

Values of the limiting dilution enthalpies of transfer ($\Delta_{\text{tr}}\Delta_{\text{dil}}H^0$) of the aqueous amino acids from water to 1 mol·dm^{-3} and 2 mol·dm^{-3} sorbitol, presented in Table 9, were calculated as follows:

$$\Delta_{\text{tr}}\Delta_{\text{dil}}H^0[\text{water to sorbitol (aq)}] = \Delta_{\text{dil}}H^0[\text{sorbitol(aq)}] - \Delta_{\text{dil}}H^0[\text{water}] \tag{19}$$

In the ternary systems under investigation (amino acids, sorbitol and water), there are three main kinds of interactions that will contribute to the value of$\Delta_{\text{tr}}\Delta_{\text{dil}}H^0$. First are interactions of the polar groups of amino acid and sorbitol that give exothermic contributions to the enthalpy of transfer. The second type of interaction consists of hydrophobic hydration around the hydrophobic/nonpolar groups of the amino acids imparting endothermic contributions to the enthalpy transfer. For the third type of interaction, the molecules also interact with each other with participation of the solvent molecules of the hydrating layer that surrounds the polar and ionic groups.

In an aqueous solution, the molecules interact with each another through participation of the solvent molecules that surround the polar and ionic groups with a hydrating layer. For direct interaction to occur, some water molecules need to be removed from the hydration shells of the interacting molecules. The removal of water molecules from the hydration shell of the polar head groups of the amino acid results is an endothermic effect. For amino acids with alkyl side chains, this process is superimposed with the additional effect

Table 8 ITC data for the titration of amino acids at various concentrations of sorbitol

Amino acid	q				
$10^3 m$ (mol·kg^{-1})	m (mol·kg^{-1})		Amino acid	m (mol·kg^{-1})	
	1.0	2.0	$10^3 m$ (mol·kg^{-1})	1.0	2.0
Glycine			Alanine		
1.40	–735	–1990	1.40	–149	–1627
2.79	–726	–1971	2.79	–149	–1592
4.17	–716	–1971	4.17	–143	–1563
5.55	–707	–1964	5.55	–152	–1533
6.91	–698	–1956	6.91	–146	–1501
8.26	–689	–1947	8.26	–145	–1469
9.61	–681	–1938	9.61	–142	–1444
10.94	–673	–1924	10.94	–144	–1413
12.26	–664	–1914	12.26	–142	–1386
13.58	–649	–1906	13.58	–136	–1360
14.88	–641	–1895	14.88	–135	–1332
16.	–634	–1889	16.18	–133	–1308
17.46	–625	–1885	17.46	–133	–1283
18.73	–615	–1876	18.73	–132	–1259
20.00	–608	–1864	20.00	–129	–1236
21.25	–604	–1861	21.25	–126	–1212
22.50	–595	–1862	22.50	–123	–1195
23.73	–586	–1854	23.73	–123	–1167
24.96	–575	–1845	24.96	–123	–1146
26.17	–571	–1826	26.17	–130	–1127
27.37	–571	–1819	27.37	–130	–1105
28.57	–559	–1812	28.57	–103	–1086
29.75	–546	–1797	29.75	–118	–1065
α-Amino butyric acid			Valine		
1.40	18	–832	0.35	28	
2.79	20	–813	0.69	33	
4.17	21	–794	1.04	68	
5.55	21	–777	1.38	79	–3987
6.91	22	–751	1.72	103	–3931
8.26	23	–739	2.06	112	–3862
9.61	24	–722	2.40	116	–3804
10.94	25	–704	2.73	123	–3739
12.26	26	–689	3.06	127	–3692
13.58	25	–672	3.39	134	–3630
14.88	25	–656	3.72	136	–3670
16.18	27	–642	4.04	147	–3528
17.46	27	–626	4.36	150	–3467
18.73	28	–611	4.68	154	–3405
20.00	28	–598	5.00	158	–3309
21.25	29	–584	5.31	155	–3249

Table 8 (*Continued*)

Amino acid	$\Delta_{dil}H^{\infty}$ (J·mol^{-1})				
10^3m (mol·kg^{-1})	m (mol·kg^{-1})		Amino acid	m (mol·kg^{-1})	
	1.0	2.0	10^3m (mol·kg^{-1})	1.0	2.0
22.50	29	–572	5.62	163	–3202
23.73	30	–556	5.93	165	–3156
24.96	30	–544	6.24	167	–3103
26.17	30	–532	6.54	170	–3091
27.37	30	–519	6.84	171	–3061
28.57	31	–507	7.14	177	–3008
29.75	31	–495			
30.93	31	–484			
Leucine					
1.04	201				
1.38	218				
1.72	237	–3499			
2.06	254	–3439			
2.40	271	–3363			
2.73	275	–3313			
3.06	279	–3263			
3.39	286	–3217			
3.72	287	–3154			
4.04	288	–3100			
4.36	292	–3062			
4.68	294	–3012			
5.00	294	–2953			
5.31	298	–2903			
5.62	298	–2857			
5.93	302	–2803			
6.24	303	–2761			
6.54	303	–2723			
6.84	303	–2677			
7.14	302	–2617			
7.43	304	–2568			

of hydrophobic hydration, causing a reinforcement of hydrogen bonds between the water molecules surrounding these alkyl groups. Studies of the structure of water in amino acid solutions indicate that the water–water hydrogen bonds in the zones surrounding the alkyl chains are stronger than those in bulk water and, as the size of the side chain increases, the size of the clathrate around the hydrophobic side chains also increases. This effect gets transferred, due to cooperation of hydrogen bonds, onto the waters in the hydration layers of the zwitterions, which reinforces the interactions between the water molecules and the polar groups of the amino acids. Thus, the exothermic effect from direct interactions between the polar groups of sorbitol is weakened and sometimes dominated by the endothermic effects of partial dehydration of the solvating layers of the interacting molecules.

Table 9 ITC results for the interaction of amino acids with various concentration of sorbitol

Amino acid	Water	q		$\Delta_{tr}\Delta_{dil}H^0$ (J·mol^{-1})	
		Sorbitol (mol·kg^{-1})		Water to sorbitol (mol·kg^{-1})	
		1.0	2.0	1.0	2.0
Glycine	87.2 (3.0)	–742.5(1.5)	–1991.5 (2.3)	829.7 (1.0)	–2078.8 (0.5)
Alanine	–45.6 (2.0)	–153.6(1.8)	–1603.9 (4.0)	–108.0 (0.1)	–1558.3 (1.4)
α-Amino-n-butyric acid	–112.2 (4.0)	19.8 (1.0)	–867.2(2.5)	132.0 (2.1)	–735.2 (1.0)
Valine	49.4 (3.0)	63.2 (5.4)	–4530.6 (7.0)	112.6 (1.7)	–4417.4 (2.8)
Leucine	–67.0 (4.0)	228.1 (6.3)	–4078.4 (6.3)	161.8 (1.6)	–3916.7 (1.6)

The enthalpic interactions of sorbitol with the glycine molecule (Gly) determined in aqueous solutions, in which –H is a side substituent, yield a negative value, indicating that the exothermic effect of direct interactions of glycine zwitterions with sorbitol dominates over the endothermic hydration effect. The replacement of the C^{α} in the amino acid molecule with an alkyl substituent in the cases of L-alanine, DL-α-aminobutyric acid, L-valine and L-leucine enhances the endothermic effect connected with partial dehydration and is intensified by hydrophobic hydration of the nonpolar alkyl groups.

A comparison of $\Delta_{tr}V^0_{2,m}$ and $\Delta_{tr}H^0$ from water to 1 mol·kg^{-1} sorbitol and to 2 mol·kg^{-1} sorbitol solutions suggests that the extent of hydrophilic interactions, in general, increases upon increasing the sorbitol concentration. These hydrophilic interactions involve the zwitterions and the –OH groups of the sorbitol. As a result of these hydrophilic interactions between the amino acids and the cosolutes, the electrostricted water is excluded to the bulk solvent, thereby shifting the monomer $\rightleftharpoons$ tetrahedral hydrogen bonded water network structure equilibrium to the right. These results also correlate well with the isentropic compressibility transfer values that are more positive in 2 mol·kg^{-1} sorbitol compared to the corresponding values in 1 mol·kg^{-1} sorbitol.

4 Conclusions

A combination of densimetry, compressibility measurements and isothermal titration calorimetry has helped in understanding the fine details of the interactions of a homologous series of amino acids with sorbitol (which is known to be a protein stabilizer). In addition to a quantitative knowledge of the contribution of various constituent groups of the amino acids to the studied thermodynamics properties, the values of $\Delta_{tr}V^0_{2,m}$, $\Delta_{tr}K^0_{S,2,m}$ and $\Delta_{tr}\Delta_{dil}H^0$ permit an understanding of the type of interactions that occur between the amino acids and aqueous sorbitol. Correlation of these results suggest enhancement of the hydrophilic/polar group interactions operating in ternary systems of amino acid, water and sorbitol.

References

1. Ou, W.B., Park, Y.D., Zhou, H.M.: Effect of osmolytes as folding aids on creatine kinase refolding pathway. Int. J. Biochem. Cell Biol. **34**, 136–147 (2002)
2. Dill, K.A.: Dominant forces in protein folding. Biochemistry **29**, 7133–7155 (1990)
3. Bolen, D.W.: Effects of naturally occurring osmolytes on protein stability and solubility: issues important in protein crystallization. Methods **34**, 312–322 (2004)

4. Gekko, K., Timasheff, S.N.: Mechanism of protein stabilization by glycerol: preferential hydration in glycerol–water mixtures. Biochemistry **20**, 4667–4676 (1981)
5. Petersen, S.B., Jonson, V., Fojan, P., Wimmer, R., Pedersen, S.: Sorbitol prevents the self-aggregation of unfolded lysozyme leading to an up to 13 °C stabilisation of the folded form. J. Biotechnol. **114**, 269–278 (2004)
6. Wimmer, R., Olsson, M., Petersen, M., Neves, N., Hatti-Kaul, R., Petersen, S.B., Muller, N.: Towards a molecular level understanding of protein stabilization: the interaction between lysozyme and sorbitol. J. Biotechnol **55**, 85–100 (1997)
7. Timasheff, S.N.: The control of protein stability and association by weak interactions with water. How do solvents affect these processes? Ann. Rev. Biophys. Biomol. Struct. **22**, 67–97 (1993)
8. Archer, D.G.: Thermodynamics properties of the NaCl + H_2O system. II. Thermodynamics properties of NaCl(aq), NaCl. H_2O(cr), and phase equilibria. J. Phys. Chem. Ref. Data **21**, 793–829 (1992)
9. Kikuchi, M., Sakurai, M., Nitta, K.: Partial molar volumes and adiabatic compressibilities of amino acids in dilute aqueous solutions at 5, 15, 25, 35, and 45 °C. J. Chem. Eng. Data **40**, 935–942 (1995)
10. Kharakoz, D.P.: Volumetric properties of proteins and their analogues in diluted water solutions. 2. Partial adiabatic compressibilities of amino acids at 15–70 °C. J. Phys. Chem. **95**, 5634–5642 (1991)
11. Desnoyers, J.E.: Structural effects in aqueous solutions: a thermodynamic approach. Pure Appl. Chem. **54**, 1469–1478 (1982)
12. Hedwig, G.R., Reading, J.F., Lilley, T.H.: Aqueous solutions containing amino acids and peptides: Part 27—Partial molal heat capacities and partial molar volume of some N-acetyl amino acids, some N-acetyl peptides and two peptides at 25 °C. J. Chem. Soc., Faraday Trans. **87**, 1751–1758 (1991)
13. Singh, S.K., Kundu, A., Kishore, N.: Interaction of some amino acids and glycine peptides with aqueous sodium dodecylsulfate and cetyltrimethyl ammonium bromide at $T = 298.15$ K: A volumetric approach. J. Chem. Thermodyn. **36**, 7–16 (2004)
14. Wadi, R.K., Goyal, R.K.: Temperature dependence of apparent molar volumes and viscosity B-coefficients of amino acids in aqueous potassium thiocyanate solutions from 15 to 35 °C. J. Solution Chem. **21**, 163–170 (1992)
15. Wang, J., Yau, Z., Zhuo, K., Liu, J.: Partial molar volumes of some α-amino acids in aqueous sodium acetate solutions at 308.15 K. Biophys. Chem. **80**, 179–188 (1999)
16. Hakin, A.W., Duke, M.M., Marty, J.L., Presuss, K.E.: Some thermodynamic properties of aqueous amino acid systems at 288.15, 298.15, 313.15 and 328.15 K: group additivity analyses of standard-state volumes and heat capacities. J. Chem. Soc., Faraday Trans. **90**, 2027–2035 (1994)
17. Hakin, A.W., Duke, M.M., Groft, L.L., Marty, J.L., Rashfeldt, M.L.: Calorimetric investigations of aqueous amino acid and dipeptide systems from 288.15 to 328.15 K. Can. J. Chem. **73**, 725–734 (1995)
18. Singh, S.K., Kishore, N.: Partial molar volume of transfer of some amino acids and peptides from water to 1 mol·dm^{-3} aqueous sodium acetate, sodium sulfate, and sodium thiocyanate at 25 °C, and correlation of the transfer parameters to the thermal stability of hen egg white lysozyme and α-lactalbumin in the presence of these salts. J. Solution Chem. **132**, 117–135 (2003)
19. Franks, F., Quickenden, M.A., Reid, D.S., Watson, B.: Calorimetric and volumetric studies of dilute aqueous solutions of cyclic ethers derivatives. Trans. Faraday Soc. **66**, 582–589 (1970)
20. Millero, F.J., Surdo, A.L., Shin, C.: The apparent molar volumes and adiabatic compressibilities of aqueous amino acids at 25 °C. J. Phys. Chem. **82**, 784–792 (1978)
21. Berline, E., Pallansh, M.J.: Densities of several proteins and L-amino acids in dry state. J. Phys. Chem. **72**, 1887–1889 (1968)
22. Teresawa, S., Itsuki, H., Arakawa, S.: Contribution of hydrogen bonds to the partial molar volumes of nonionic solutes in water. J. Phys. Chem. **79**, 2345–2351 (1975)
23. Bondi, A.: van der Waals volumes and radii. J. Phys. Chem. **68**, 441–451 (1964)
24. Bondi, A.: Free volumes and free rotation in simple liquids and liquid saturated hydrocarbons. J. Phys. Chem. **58**, 929–939 (1954)
25. Shahidi, F., Farell, P.G., Edwards, J.T.: Partial molar volume of organic compounds in water carbohydrates. J. Solution Chem. **5**, 807–816 (1976)
26. Banipal, T.R., Sehgal, G.: Partial molal adiabatic compressibilities of transfer of some amino acids and peptides from water to aqueous sodium chloride and aqueous glucose solutions. Thermochem. Acta **262**, 175–183 (1995)
27. Bhat, R., Kishore, N., Ahluwalia, J.C.: Thermodynamics of some amino acids and peptides from water to aqueous glucose and sucrose solutions at 298.15 K. J. Chem. Soc., Faraday Trans. I **84**, 2651–2665 (1988)
28. Sharma, R., Kishore, N.: Interaction of some amino acids with aqueous osmoprotectant proline at 298.15 K. J. Solution Chem. **35**, 231–249 (2006)
29. Jasra, R.V., Ahluwalia, J.C.: Thermodynamics of transfer of sorbitol and mannitol from water to aqueous solutions of urea, guanidine hydrochloride and sodium chloride. J. Chem. Soc., Faraday Trans. **1**, 1677–1687 (1982)

30. Jasra, R.V., Ahluwalia, J.C.: Enthalpies, enthalpies of solution, partial molal heat capacities and apparent molal volumes of sugars and polyols in water. J. Solution Chem. **11**, 325–338 (1982)
31. Xu, L., Wang, X., Lin, R., Sun, D.: Enthalpies of dilution of glycine and L-alanine in aqueous 1-propanol solutions at $T = 298.15$ K. J. Chem. Thermodyn. **37**, 371–375 (2005)